SymNMF Final Project: Forum Instructions

Software Project 0368-2161, 2025/2026. A summary of every answer the course staff gave in the Moodle project forum, collected on 29 Sep 2026.

Staff wrote that answers in the forum **should be considered instructions**, so they apply together with the project PDF. Each box below is one instruction, followed by the forum thread and date it came from. Threads that were only office-hours announcements are left out.

1. Input validation and error messages

Validate the command line arguments like in the previous homework, in both C and Python. k must satisfy **1 < k < N**. You can assume the contents of the input files are valid.

Thread: Input validations, 16 Aug 2026

For invalid arguments, print the error messages **as in the previous homework**, for example **"Incorrect number of clusters!"** for an invalid k. Staff: the answers in the forum should be considered instructions.

Thread: Error message and kmeans questions, 22 and 23 Aug 2026

k always needs to be validated, also when the goal is sym, ddg or norm. An invalid goal prints **"An Error Has Occurred"**.

Thread: Questions about k and error handeling, 23 Aug 2026

All three arguments are always given in Python, and two in C. The program must not ignore a missing argument, even one it does not need. It should print "An Error Has Occurred" and exit.

Thread: Questions about Multiple Topics, answer 1, 24 Sep 2026

k written as a whole decimal: the previous homework answers apply. From the HW1 forum: **3, 3.0, 3.00, 3.000 and 3. are valid** and mean 3. **3.5, 3.01, -3 and abc are invalid**. 3.00000000000000000000000000000000000001 is also invalid because it is not a natural number.

Threads: Is k written as "3.0" valid?, 22 Aug 2026, which refers to the HW1 thread Validation of K and iter as Natural Numbers, 8 to 24 Jun 2026

2. Input file

The data file will be in the same format as in the previous homework.

Thread: Assumptions about input's last row, 17 Aug 2026

You can assume the input file is valid. There is no need to reject a file because its name does not end with .txt.

Thread: Input file extension, 23 Aug 2026

The input file will not be empty.

Thread: Empty input file, 6 Sep 2026

One dimensional points, d = 1, must be supported.

Thread: One-dimensional input, 27 Sep 2026

3. The SymNMF algorithm

Add a small epsilon, **1e-6 or smaller**, to the denominator of the H update in section 1.4.2 to avoid division by zero. It is better to **always** add it, not only when the denominator is 0. A smaller value like 1e-12 is fine. Expected numerical differences are up to 0.0001. If a test fails only because of the added 1e-6, an appeal will be accepted.

Thread: division by zero in update H, 2 Sep to 24 Sep 2026

Call **np.random.seed(1234)** at the beginning of the block that handles the symnmf algorithm.

Thread: Questions about Multiple Topics, answer 4, 24 Sep 2026

A zero degree in D, which would make D^-1/2 undefined, should not happen with the test inputs. You **can** add a small epsilon to the denominator there too.

Thread: Questions about Multiple Topics, answer 3, 24 Sep 2026

4. C extension and program structure

The C extension module is named **symnmfmodule**. Naming it symnmf may pass the tests but might cause problems in analysis.py.

Threads: C extension module name, and Error message and kmeans questions, 22 Aug 2026

Only the symnmf() wrapper may take extra arguments. sym(), ddg() and norm() get only the data points.

Thread: Wrappers and Pandas, answer 1, 2 Sep 2026

For goal symnmf, the Python code should build W by calling the relevant C module function, norm(), and then use W for the initial H and pass it to symnmf().

Thread: Wrappers and Pandas, 6 Sep 2026

ddg() returns the diagonal degree matrix as described in sections 2.1.1, 2.1.3 and 2.2.1. Staff did not allow returning only the diagonal values.

Thread: Matrix returned from ddg method by C module, 20 Aug 2026

The C program must be standalone with a main function, because it cannot be compiled without one.

Thread: Questions about Multiple Topics, answer 5, 24 Sep 2026

5. analysis.py and the kmeans code from HW1

You need to adjust the HW1 code to the project instructions, for example the function length limit and the epsilon and max_iter values.

Thread: question about analysis.py, 22 Aug 2026

For kmeans use the project epsilon, **1e-4**, and handle an empty cluster **as in HW1**.

Thread: Error message and kmeans questions, answer 3, 22 Aug 2026

Keeping the HW1 code with lists or rewriting it with numpy are both valid.

Thread: changes to hw1, 27 Aug 2026

analysis.py must make sure **silhouette_score does not throw an error**, for example when all points end up in one cluster. Check before calling it, catch the error, or both. If an error happens, print it and exit.

Thread: One-dimensional input, 27 Sep 2026

scikit-learn is not in the course Docker image. You need to install it.

Thread: scikit-learn in the course Docker, 19 Sep 2026

6. Code style and documentation

Every function, including main, is limited to **40 lines including documentation**. Blank lines are not counted. A function prototype does not count. A signature split over two lines counts as two lines.

Thread: Function length limit, 19 and 20 Aug 2026

Documentation is required for the manual test. For each function explain what it does, its input and output, and any non-trivial part of the code.

Thread: Documentation, 26 Aug 2026

Document the C functions in **both** the header file and the .c file.

Thread: Questions about Multiple Topics, answer 2, 24 Sep 2026

Allowed libraries: **math.h**, **stdbool.h**, **string.h** in C, and **pandas** in Python.

Threads: library math, 10 Sep; Using libraries in C, 23 Aug; Wrappers and Pandas, 2 Sep 2026

7. Submission

The default is submitting in pairs. Some students got permission to submit alone.

Thread: submission, 7 Sep 2026

A pair submits **one** file with both IDs, id1_id2_project.tar.gz, as in HW1 and HW2. For an individual submission id2 is 111111111, as written in the project instructions.

Thread: Submission in couples, 29 Sep 2026

No test files are provided for the project. Create your own.

Thread: test files, 24 Aug 2026